

$$1. 1) \vec{q} = -k \nabla T = -0.292 \frac{\text{W}}{\text{m} \cdot \text{K}} \cdot (3\vec{e}_x + 4\vec{e}_y + 5\vec{e}_z) \frac{\text{K}}{\text{m}} \\ = -0.876\vec{e}_x - 1.168\vec{e}_y - 1.46\vec{e}_z \left(\frac{\text{W}}{\text{m}^2} \right)$$

$$2) \vec{q} = -\bar{k} \nabla T \\ = - \begin{bmatrix} 10 & 5 & 0.38 \\ 5 & 10 & 5 \\ 0.38 & 5 & 10 \end{bmatrix} \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$$

$$= - (30 + 20 + 1.9, 15 + 40 + 25, 1.14 + 20 + 50)$$

$$= - (51.9, 80, 71.14)$$

$$= -51.9\vec{e}_x - 80\vec{e}_y - 71.14\vec{e}_z \left(\frac{\text{W}}{\text{m}^2} \right)$$

$$2. \frac{Q}{t} = \frac{kA \Delta T}{L}$$

$$k = \frac{Q}{t} \cdot \frac{L}{A \Delta T} = 3 \text{ W} \times \frac{0.5 \text{ cm} \times \frac{1 \text{ m}}{100 \text{ cm}}}{5 \text{ K} \times 10 \text{ cm}^2 \times \frac{1 \text{ m}^2}{(100 \text{ cm})^2}} = 0.3 \cdot \frac{\text{W}}{\text{m} \cdot \text{K}}$$